# Vishal Ravi Muthaiah

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## EDUCATION

**University at Buffalo, SUNY** — *Buffalo, NY*
*Master of Science in Data Science*

**Dayananda Sagar College of Engineering** — *Bengaluru, India*
*Bachelor of Engineering in Computer Science*

## PROJECTS

**Multi-Agent RL Movie Recommendation (QMIX/MAPPO)** — *Reinforcement Learning / Recommender Systems*
- Built an 11-agent cooperative recommendation system using QMIX, MAPPO, and DQN with persona-based reward shaping across 5 user archetypes, where MAPPO converged 13x faster than DQN and outperformed SVD-based collaborative filtering on cold-start users with no rating history while maintaining warm-start Hit@10 parity with traditional baselines.
- Designed a mixed reward function combining preference similarity, genre relevance, and rating quality with a cooperative coverage bonus and redundancy penalty, improving team reward by 12.6% and scaling catalog coverage 4.6x as the number of agents grew from 2 to 10.
- Deployed a FastAPI + Next.js interface with Borda rank fusion across all 11 agents, persona- matched onboarding, and per-movie explainability scores, evaluated using Hit@10, NDCG@10, ILD, and paired bootstrap significance tests across 5,000 resamples.

**CT Scan Tampering & Lesion Detection** — *Deep Learning / Medical Imaging*
- Fine-tuned EfficientNet-B0 and ResNet18 in PyTorch for binary lesion classification, reaching 0.80 validation accuracy on held-out patient scans, with a CUDA/cuDNN preprocessing pipeline running roughly 5x faster than the CPU-only baseline.
- Applied Grad-CAM to verify the model focused on clinically relevant regions, improving interpretability and building stakeholder trust for downstream review by medical practitioners.

**Zone of Inhibition Detection** — *Published in IJFMR* – Read Paper
- Built an automated antibiotic susceptibility testing system using YOLOv5, OpenCV, and TensorFlow, reaching 95% accuracy on resistance classification across a curated dataset of lab plate images.
- Improved zone measurement accuracy by 30% through custom image preprocessing and segmentation, and trained detection models with hyperparameter tuning to handle variable lighting and plate conditions.
- Deployed the full ML workflow behind REST APIs for automated inference, with MySQL-backed storage for results and performance monitoring dashboards used by lab technicians in daily operations.

**Neural Network from Scratch (MNIST)** — *NumPy / Deep Learning Fundamentals*
- Implemented a one-hidden-layer feedforward neural network in pure NumPy for binary MNIST classification (even vs. odd digits), with hand-derived backpropagation and full-batch gradient descent without any autodiff framework.
- Verified analytical gradients against centered finite differences in float64, achieving differences on the order of 1e-10 and debugging a float32 catastrophic cancellation issue during gradient checking.
- Benchmarked learning rates and hidden layer widths (64, 128, 256), reaching 97.85% test accuracy with the best configuration and analyzing convergence stability, runtime scaling, and memory usage trade-offs.

## EXPERIENCE

**L&T Technology Services** — June 2024 – July 2025
*Software Engineer* — *Bengaluru, India*
- Designed RESTful backend services in Python (Flask) and Node.js with MongoDB and SQL, handling structured product data across hundreds of part combinations for storage, retrieval, and validation across internal tools. Built a real-time Socket.io chat platform with group messaging and notifications, adopted by 100+ users and cutting cross-team response latency to under 200ms in production.
- Developed ML models for anomaly detection and predictive maintenance on sensor data, improving failure prediction accuracy by 35% over the existing rule-based system; presented bi-weekly updates to engineering leadership, translating model metrics into business impact to shape the division's AI roadmap.

**Bosch Global Software Technologies** — Aug 2023 – June 2024
*Data Science Intern* — *Bengaluru, India*
- Built a Responsible AI monitoring dashboard tracking model drift, fairness metrics, and inference latency across production ML models, giving stakeholders real-time visibility into overall model health.
- Connected NLP and ML monitoring APIs to Power BI reporting, cutting stakeholder decision turnaround by ∼25%, and translated analytical findings into recommendations for technical and non-technical audiences.

**Neuberg-Anand Diagnostic Laboratory** — Sep 2023 – Feb 2024
*Machine Learning Intern* — *Bengaluru, India*
- Automated antibiotic susceptibility measurement using Python and TensorFlow, replacing a manual lab workflow and improving measurement accuracy by 30% while cutting manual effort by ∼60% across daily testing runs; work was recognized with a Best Innovative Project award and later published in IJFMR.

## PUBLICATIONS & AWARDS

- Published *"Inhibition Zones: A Window into Antibiotic Resistance"* in the International Journal for Multidisciplinary Research (IJFMR); awarded Best Innovative Project at Project Open Day, Dayananda Sagar College of Engineering.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Java, C++, C, JavaScript, TypeScript, R, React JS, Node JS, Express JS, HTML/CSS
**ML / Generative AI:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, Keras, OpenCV, prompt engineering, RAG, embeddings, fine-tuning, Grad-CAM, Deep Learning, Reinforcement Learning, Multi-Agent RL (QMIX, MAPPO), PettingZoo, Gymnasium, reward shaping, Recommender systems, NLP, Computer Vision
**Backend & Infra:** FastAPI, Flask, Django, Node.js, Express.js, REST APIs, Docker, Git, CI/CD, AWS, Azure, Linux, CUDA, Socket.io
**Data & Analytics:** Pandas, NumPy, MongoDB, MySQL, PostgreSQL, Power BI, Tableau, Matplotlib, Seaborn, Jupyter, Excel